DATACAMP-INSPIRED PROJECT REPORT

# HR Analytics in Power BI

A major case study from DataCamp's 50-hour Data Analyst in Power BI track, exploring hiring, workforce demographics, performance, and attrition at fictional Atlas Labs.

POWER BI / DAX / DATA MODELING / HR ANALYTICS

**PROJECT REPOSITORY**

https://github.com/M-Gaber056/hr-analysis-in-powerbi

**LINKEDIN POST**

*https://lnkd.in/p/eutCFf8y*

**INSIDE THIS GUIDE**

Project overview / Data and model / Analytical workflow / Questions and interpretation / Visual design / Reproducibility

*Visual styling follows DataCamp Waffles theme tokens; this is an independent portfolio document, not an official DataCamp publication.*

01 / EXECUTIVE SUMMARY

# Project overview

This project applies Power BI to a human-resources analytics case study based on Atlas Labs, a fictional software company used in DataCamp's Case Study: HR Analytics in Power BI. It was one of the major projects in the 50-hour Data Analyst in Power BI track. The report brings employee, hiring, performance, and attrition information into a coherent analytical view that helps stakeholders explore workforce patterns.

**WORKFORCE**

Explore employee composition and job-related attributes.

**HIRING**

Review hiring activity over time and across roles.

**PERFORMANCE**

Examine review ratings, satisfaction, and related measures.

**ATTRITION**

Investigate employee exits and patterns associated with attrition.

## Objectives

- Consolidate workforce and performance information into a connected analytical model.
- Use exploratory analysis and report visuals to surface patterns in hiring, employee profiles, performance, and attrition.
- Apply DAX and a date table to support time-aware analysis.
- Present the analysis in a consistent, branded, navigable Power BI report.

## Project deliverables

- Power BI report file: CaseStudyHrAnal.pbix
- Source tables and lookup data in the Datasets folder.
- A custom Power BI theme in Theme.json and a DAX date-table definition in DimDate.txt.

**PROJECT CONTEXT**

The work is a portfolio case study using a fictional company dataset. It is intended to demonstrate analytical workflow and Power BI reporting skills, not to make claims about a real employer or workforce.

02 / SOURCE ARTIFACTS

# Data and model

The repository includes the report file, employee and review data, small lookup tables, a date-table definition, and the custom report theme. Together, these files support workforce exploration and performance/attrition analysis.

| REPOSITORY FILE | ROLE | WHAT IT CONTRIBUTES |
|---|---|---|
| **CaseStudyHrAnal.pbix** | Power BI report | The main interactive report and its model, measures, and visuals. |
| **Employee.csv** | Employee attributes | Employee ID, demographics, department, role, salary, hire date, attrition, and tenure-related fields. |
| **PerformanceRating.csv** | Review records | Review date, satisfaction measures, training opportunities, work-life balance, and self/manager ratings. |
| **EducationLevel.csv** | Lookup table | Maps education-level IDs to readable education categories. |
| **RatingLevel.csv / SatisfiedLevel.csv** | Lookup tables | Provide labels for rating and satisfaction scale values. |
| **DimDate.txt** | DAX date table | Defines a calendar table based on hire-date boundaries and sets April as the fiscal-year start month. |
| **Theme.json** | Visual theme | Defines the report name and a DataCamp-aligned green/navy color palette. |
| **Metadata - Case Study_HR Analytics in Power BI.pdf** | Reference material | Course-provided metadata reference for understanding the tables and fields. |

## Conceptual relationships

| EMPLOYEE | EMPLOYEEID → EMPLOYEEID | PERFORMANCERATING |
|---|---|---|
| Employee profile, role, hire date, attrition, tenure | Connect people to dated review records; use dimensions for readable categories | Performance reviews, satisfaction, training, ratings |

The data files show EmployeeID in both employee and performance records, making it the key field for connecting a person's profile to review history. The separate lookup tables support clearer labels, while the date table provides a consistent time context.

03 / IMPLEMENTATION

# Analytical workflow

The work follows the report-building sequence taught in the DataCamp case study: load and relate the data, explore the business questions, build the visuals, then improve navigation and the branded layout.

**01** **Import and inspect**
Load the employee, performance, and lookup tables. Review fields and data types before building visuals.

**02** **Model the data**
Connect employee and review records through EmployeeID; add lookup dimensions and a reusable DAX date table.

**03** **Explore the workforce**
Analyze hiring activity, departments and roles, and the composition of the workforce.

**04** **Review performance**
Compare performance ratings alongside satisfaction, training-opportunity, and work-life-balance fields.

**05** **Investigate attrition**
Use employee attributes and tenure-related fields to explore how attrition varies across workforce groups.

**06** **Polish the report**
Apply the custom HR Analytics theme, keep visual formatting consistent, and make report navigation intuitive.

## Tools and techniques

| | |
|---|---|
| POWER BI DESKTOP | Report authoring and interactive visuals |
| DAX | Measures and calendar table logic |
| DATA MODELING | Fact/detail records and lookup tables |
| EXPLORATORY ANALYSIS | Comparisons across time and employee groups |
| THEME.JSON | Reusable visual styling |

04 / ANALYTICAL FRAMEWORK

# Questions the report helps explore

Rather than treating HR metrics as isolated numbers, the report brings several complementary views together. The questions below describe the intended analytical lens; consult the PBIX file for the exact measure values and visuals in the current report version.

01

### Hiring and workforce growth

How has hiring changed over time? Which departments and job roles account for different parts of the workforce?

*Useful views: hiring timeline, department/role breakdowns, and employee counts over time.*

02

### Employee demographics

What does the workforce look like across age, gender, ethnicity, education, marital status, and role?

*Useful views: distributions and comparisons across selected employee attributes.*

03

### Performance and experience

How do self-ratings and manager ratings compare? How are satisfaction, training opportunities, and work-life balance represented in review records?

*Useful views: rating distributions and comparisons by review or employee group.*

04

### Attrition patterns

How does attrition vary across departments, roles, tenure, overtime, travel, or other available attributes?

*Useful views: attrition rate and grouped comparisons, interpreted as descriptive associations.*

**INTERPRETATION PRINCIPLE**

Differences between employee groups are signals for further investigation, not proof that one factor causes attrition or performance outcomes. Read each comparison alongside sample sizes, definitions, and the context of the fictional dataset.

05 / DATA STORYTELLING

# Visual design and interpretation

The repository includes a Power BI theme named “HR Analytics.” Its first data colors align with the distinctive DataCamp green and deep navy. This document uses the same design logic: strong dark text, vivid green accents, calm neutral surfaces, and clear typographic hierarchy.

| TOKEN | HEX | ROLE | SWATCH |
|---|---|---|---|
| **Brand green** | #03EF62 | Primary accent and emphasis | |
| **Deep navy** | #05192D | Headings, dark panels, high-contrast text | |
| **Darker green** | #00C74E | Secondary accent | |
| **Light green** | #65FF8F | Light accent / highlight | |
| **Light neutral** | #F7F7FC | Soft page or card background | |

Typography: the DataCamp Waffles design-token guidance specifies Studio-Feixen-Sans with Arial as a fallback for regular text. This report uses Arial for broad compatibility while following the brand palette and a light-theme hierarchy.

## Responsible interpretation

- The case study uses a fictional company dataset; do not present its patterns as evidence about any real organization.
- Attrition and performance comparisons are descriptive. They can identify relationships worth investigating but do not establish causation.
- Demographic measures can be sensitive. Use neutral wording and avoid ranking groups as inherently better or worse.
- If the underlying PBIX changes, update any copied screenshots or metric values in the documentation so the report remains consistent.

**ABOUT THIS DOCUMENT**

This is an independent portfolio write-up designed to explain the project. It is not produced, reviewed, or endorsed by DataCamp.

06 / PROJECT HANDOFF

# Review, reproduce, and share

## How to review the project

**01** **Open the report**
Download or open CaseStudyHrAnal.pbix from the repository using Power BI Desktop.

**02** **Review the supporting data**
Inspect the files under Datasets to understand the employee, review, and lookup data used by the report.

**03** **Check the theme and date logic**
Open Theme.json for the visual palette and DimDate.txt for the DAX calendar-table definition.

**04** **Validate the current numbers**
Use the report itself as the source of truth for KPI values, filters, chart details, and any findings quoted publicly.

## Project links

**GitHub repository:** github.com/M-Gaber056/hr-analysis-in-powerbi
**DataCamp case study:** Case Study: HR Analytics in Power BI
**DataCamp Waffles themes:** Waffles theme guidance / brand tokens
**DataCamp Waffles design tokens:** Typography and theme token reference

## Published LinkedIn post

PUBLISHED POST
Published LinkedIn post: https://lnkd.in/p/eutCFf8y
This post shares the HR Analytics case study and links to the project repository.

Portfolio context: This case study was one of the major projects completed as part of DataCamp's 50-hour Data Analyst in Power BI track. It applies Power BI, DAX, data modeling, and report theming to workforce data from fictional Atlas Labs.

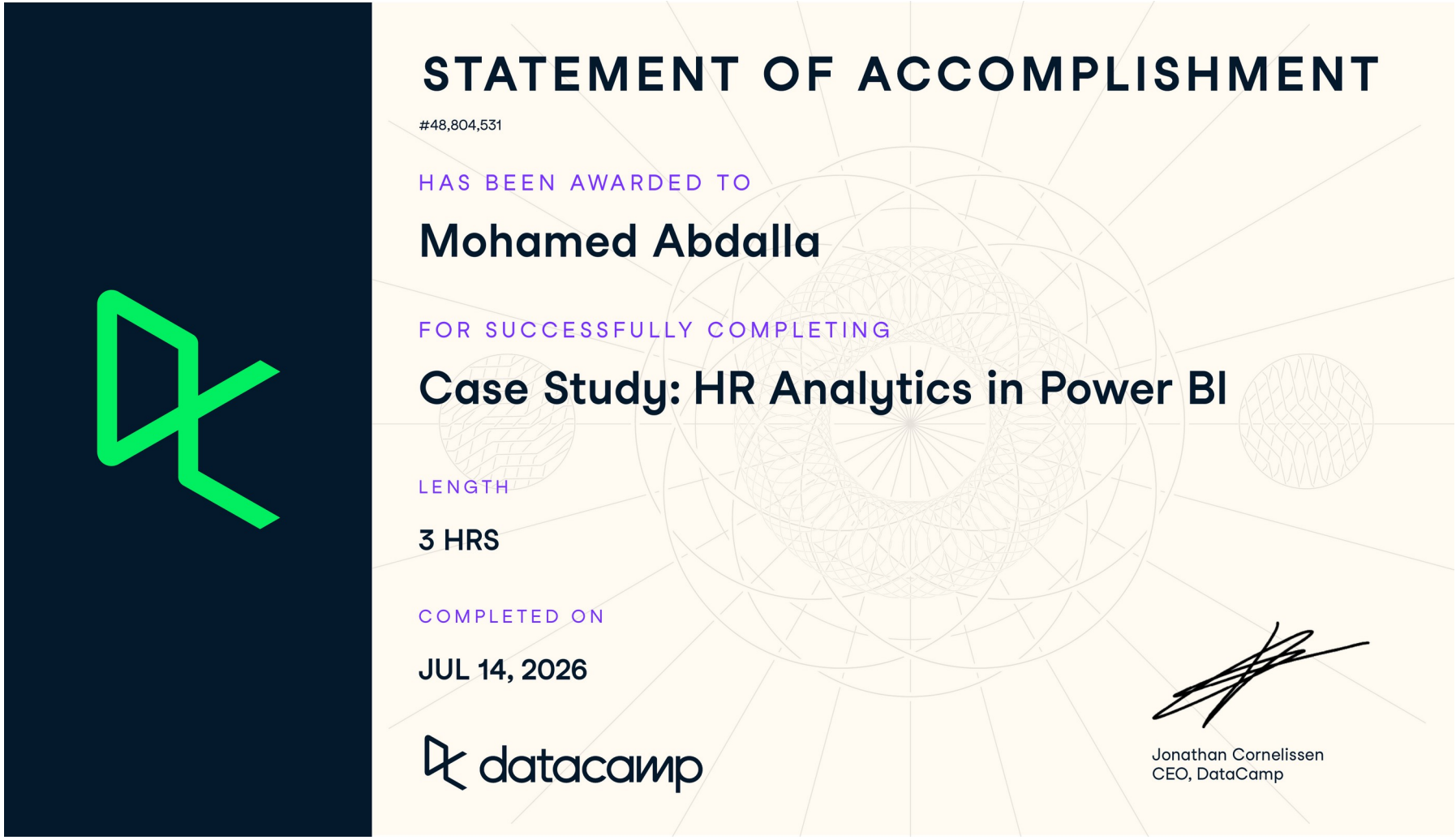
REFERENCES AND PROJECT BASIS

- Repository artifacts: PBIX, datasets, DimDate.txt, Theme.json, and metadata PDF. Source
- DataCamp course scope: data modeling and EDA, demographics and performance, attrition, report design, and theming. Source
- Waffles design tokens: brand colors, base themes, and font guidance. Source

CERTIFICATE 01 / CASE STUDY COMPLETION

## Case Study: HR Analytics in Power BI

This DataCamp Statement of Accomplishment confirms successful completion of the 3-hour HR Analytics in Power BI case study on July 14, 2026. It recognizes focused, applied work using Power BI to explore HR data and develop an analytical report, complementing the project documentation presented in this portfolio.



CERTIFICATE 02 / LEARNING TRACK COMPLETION

## Data Analyst in Power BI

This certificate confirms completion of DataCamp's 50-hour Data Analyst in Power BI track on August 3, 2026. It represents the broader learning path in Power BI analytics; the HR Analytics case study documented in this portfolio was one of the major projects completed during the track.

